MILITARY SPECIFICATION
INITIATOR, PROPELLANT ACTUATED, DELAY, JAU-3/A25 and M113 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Propellant Actuated, Delay JAU-3/A25 and M113 Metal Parts Assembly. The assembly will ultimately be loaded and used in connection with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-50375	- Initiator Propellant Actuated, M104 Metal Parts Assembly

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber "O" Rings

DRAWINGS

Munitions Command

D11731621	- Initiator Propellant Actuated Metal Parts Assembly

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:34Z
Check the source to verify that this is the current version before use.

2.1 (Cont'd)

D10551447	- Initiator, Propellant Actuated Metal Parts Assembly
C9252573	- Carton, Packing, Ammunition for Initiator, XM113
C9252574	- Box, Packing Ammunition for Initiator, XM113
IEL-11731621	- Inspection Equipment List for Initiator Propellant Actuated, Delay, Metal Parts Assembly
IEL-10551447	- Inspection Equipment List for Initiator Propellant Actuated, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Mechanical Properties.-Mechanical properties required by the drawing shall be uniform throughout the part within specified tolerances.

3.3 Shear Pin Strength.-The shear pin after application of protective finish shall shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.4 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour.

3.5 Residual magnetism.-The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

3.6 Workmanship.-The requirement for workmanship is as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-The component shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation in accordance with contract requirements and shall consist of 50 shear pins from each lot of shear pins inspected by the contractor for shear pin strength, 5 assemblies and 10 sets of unassembled components for visual and dimensional examination and tests.

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.2.1.1 (Cont'd)

of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-An ammunition lot shall consist of a group of metal parts assemblies produced by a homogeneous process.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and criteria in tables I and II. For an individual defect (single characteristic) the criteria in table I shall be used. For total defects within a major or minor group, the criteria in table II shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The surface roughness specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE I

Lot Size	Sample Size	MAJOR Accept	MAJOR Reject	MINOR Accept	MINOR Reject
1- 300	100% examination	-	-	-	-
301- 1300	110	1	2	2	3
1301- 3200	150	2	3	3	4
3201- 8000	225	3	4	4	5
8001-22000	300	4	5	5	6

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:34Z
Check the source to verify that this is the current version before use.

TABLE II

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects No. Defects in Class	110	150	225	300
2-3	1	2	3	4
4-7	2	3	4	5
8-19	3	4	5	7
20-43	4	5	8	10
Over 43	6	8	11	14

Sample Size for Minor Defects No. Defects in Class	110	150	225	300
2-3	2	3	4	5
4-7	3	4	5	7
8-16	4	5	8	10
17-48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Housing, Dwg. D11731622 covering a detail of Dwg. D11731621

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of internal thread, max.	Gage
104. Major diameter of internal thread, max.	Gage
105. True position of pitch diameter of external thread with outer diameter of body, small end	Gage
106. Concentricity of center hole with center cavity	Gage
107. Concentricity of pitch diameter of internal thread, at large end with center cavity	Gage
108. Concentricity of pitch diameter of internal thread at small end with center cavity	Gage
109. Perpendicularity of base of center cavity with wall of center cavity	Gage

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2.1 (Cont'd)

	Defect	Method of Inspection
MAJOR		
110.	Perpendicularity of cavity base at large end with pitch diameter of the internal threads	Gage
111.	True position of shear pin hole with center cavity	Gage
112.	True position of wrench flats with pitch diameter of internal thread at small end	Gage
113.	Diameter of shear pin hole	Gage
114.	Diameter of shear pin hole counterbore	Gage
115.	Diameter of center hole, min.	Gage
116.	Diameter of counterbore, large end, max.	Gage
117.	Diameter of shoulder	Gage
118.	Depth of shear pin hole counterbore	Gage
119.	Depth of shear pin hole	Gage
120.	Length of internal threads both ends, min.	Gage
121.	Width of thread undercut	Gage
122.	Location of shear pin hole	Gage
123.	Thickness of wall between cavities	Gage
124.	Depth of cavity, large end	Gage
125.	Diameter of center cavity	Gage
126.	Diameter of counterbore small end	Gage
127.	Protective finish inadequate or defective	Visual
128.	Surface finish improper in center cavity and chamfer at mouth, each end	Visual
129.	Burr or sharp corner	Visual
130.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of internal thread, min.	Gage
202.	Pitch diameter of external thread, max.	Gage
203.	Diameter of center hole, max.	Gage
204.	Diameter of small end	Gage
205.	Minor diameter of cavity, large end	Gage
206.	Minor diameter of cavity, small end	Gage
207.	Diameter of counterbore, large end, min.	Gage
208.	Diameter of counterbore, small end	Gage
209.	Diameter of chamfer, large end	Gage
210.	Diameter of chamfer, small end	Gage
211.	Diameter of body, large end	Gage

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2.1 (Cont'd)

Defect	Method of Inspection
MINOR	
212. Diameter of shear pin hole, min.	Gage
213. Diameter across wrench flats	Gage
214. Depth of counterbore, large end	Gage
215. Depth of counterbore, small end	Gage
216. Depth of bore, small end	Gage
217. Length to shoulder, small end	Gage
218. Total length	Gage
219. Thickness of shoulder	Gage
220. Location of external thread	Gage
221. Radius or chamfer missing or incorrect	Visual
222. Surface finish improper, except as otherwise classified	Visual

4.3.2.2 Pin, Shear, Dwg. B10520475, covering a detail of Dwg. D11731621

CRITICAL	
None defined	
MAJOR	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Surface finish improper	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Profile of end	Gage

4.3.2.3 Bracket, Mounting Dwg. D11731624 covering a detail of Dwg. D11731621.

CRITICAL	
None defined	

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:34Z
Check the source to verify that this is the currentversion before use.

4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of internal thread, max.	Gage
102. Major diameter of internal thread, min.	Gage
103. Diameter of mounting holes	Gage
104. Length of mounting holes centers	Gage
105. Height from base to thread hole center	Gage
106. Length from shoulder to thread hole center	Gage
107. Shoulder radius incorrect	Gage
108. Width of shoulder	Gage
109. Protective finish inadequate or defective	Visual
110. Burr or sharp corner	Visual
111. Foreign matter	Visual
MINOR	
201. Pitch diameter of internal thread, min.	Gage
202. Minor diameter of internal thread, max.	Gage
203. Total base length	Gage
204. Width of mounting surface (base)	Gage
205. Height from base to top of shoulder	Gage
206. Thickness of base	Gage
207. Length from center line to base end	Gage
208. Mounting hole location	Gage
209. Length of base relief	Gage
210. Depth of base relief	Gage
211. Base relief location incorrect	Gage
212. Angle of base relief incorrect	Gage
213. Radius, angles, or chamfers missing or incorrect	Gage
214. Surface finish improper	Visual

4.3.2.4 Pin, firing, Dwg. C10520474, covering a detail of Dwg. D11731621.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of striker with body	Gage
102. Concentricity of groove with body	Gage

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:34Z
Check the source to verify that this is the currentversion before use.

4.3.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
103. Symmetry of pin hole with body	Gage
104. Diameter of striker	Gage
105. Diameter of body	Gage
106. Diameter of groove	Gage
107. Diameter of pin hole	Gage
108. Depth of pin hole	Gage
109. Length of striker, min.	Gage
110. Location of pin hole	Gage
111. Metal defective	Gage
112. Protective finish inadequate or defective	Gage
113. Surface finish of body improper	Gage
114. Burr or sharp corner	Visual
115. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Depth of slot	Gage
203. Length of striker, max.	Gage
204. Location of groove	Gage
205. Width of slot	Gage
206. Width of groove	Gage
207. Location of slot	Gage
208. Radius or chamfer missing or incorrect	Visual
209. Surface finish improper, except as otherwise classified	Visual

4.3.2.5 Components.-Component parts of this metal parts assembly shall conform to the requirements specified in MIL-I-50375.

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Shear pin strength.-Fifty pins from each lot of shear pins tested by the contractor shall be tested for shear strength. Failure of any pin to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.3.3.2 Protective finish.-The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirement of 3.4 and the applicable specification shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.3.3 Residual magnetism.-Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.5 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 11731621 and IEL 10551447.

4.4 Test methods and procedures.

4.4.1 Shear pin strength.-After application of protective finish, the pin shall be subjected to the shear strength test to determine compliance with 3.3.

4.4.2 Protective finish.-Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.3 Residual magnetism.-The initiator shall be tested for excessive residual magnetism using an approved compass, Type AN-5766 in an area free of local magnetic effects by placing the initiator six inches from and at the same height as the compass, in a north-south heading (cap end at south) with the compass set in an east-west heading.

The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with drawings C9252573 and C9252574.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian

Army- MU

Preparing activity

Army - MU (FA)

Project No. 1377-A486

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:34Z
Check the source to verify that this is the current version before use.